CLAIMS

What is claimed is:

1. A method for dynamically composing a computer, the method comprising the steps of:
   - receiving, at a software control plane, a request for a logical computer specifying a set of hardware resources;
   - identifying available, physically disaggregated hardware components from a plurality of resource pools that satisfy the request;
   - instructing a data fabric switch to establish a plurality of communication links that interconnect the identified hardware components; and
   - provisioning the interconnected hardware components as a single logical computer.

2. The method of claim 1, further comprising the steps of:
   - receiving a signal indicating the logical computer is no longer needed;
   - instructing the data fabric switch to dissolve the communication links; and
   - returning the hardware components to the plurality of resource pools.

3. The method of claim 1, wherein the logical computer is a bare-metal computer, without a hypervisor layer between the hardware and an operating system.

4. The method of claim 1, wherein the resource pools include a pool of processors and a pool of memory modules housed in separate chassis.

5. The method of claim 1, wherein the communication links are massively parallel optical links.

6. The method of claim 5, wherein the data fabric switch is an optical data fabric switch.